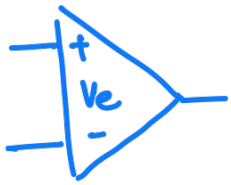
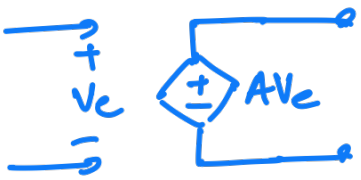
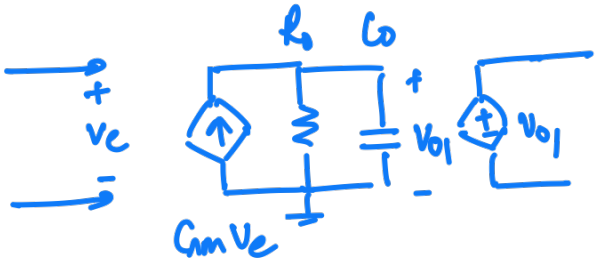
# Lecture #10



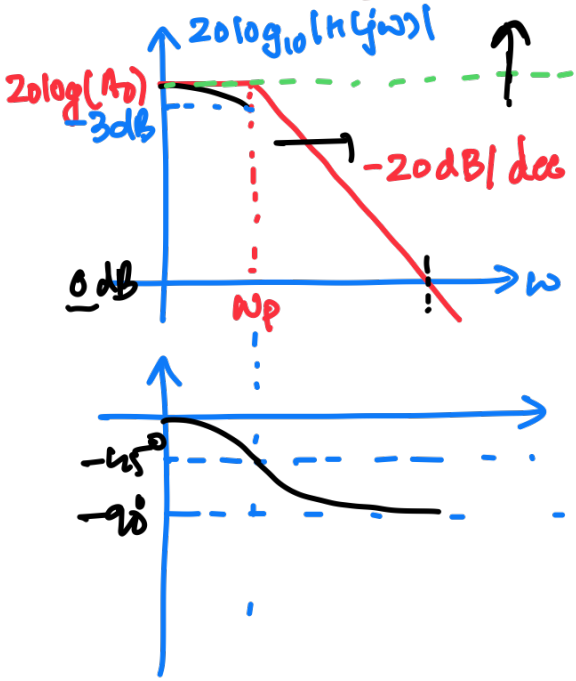




$$\frac{V_o}{V_e} = \frac{A_o}{1+s/\omega_p} = \frac{G_m R_o}{1+sR_oC_o} = H(s)$$

$A_o = G_m R_o$ (DC gain)

$\omega_p = \frac{1}{R_oC_o}$ (pole freq)

$s_1 = -\omega_p$ (complex)




$$H(j\omega) = \frac{G_m R_o}{1+j\omega R_oC_o} \checkmark$$

$$|H(j\omega)| = \frac{G_m R_o}{\sqrt{1+\omega^2 R_o^2 C_o^2}} \Big|_{\omega=\omega_p} = \frac{G_m R_o}{\sqrt{1+1}}$$

for $\omega > \omega_p$

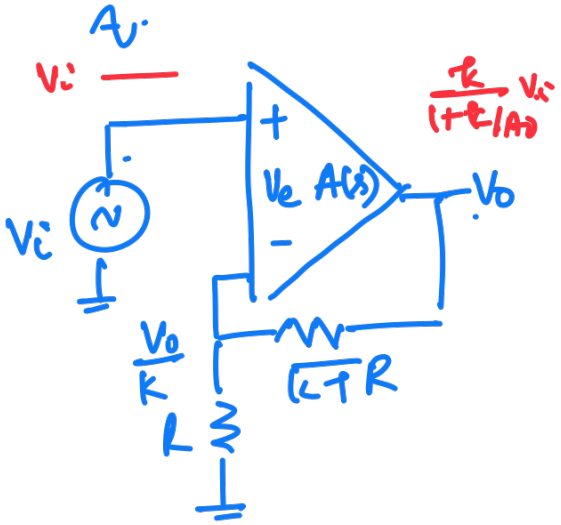
$$|H(j\omega)| \simeq \frac{A_o}{\omega/\omega_p}$$

$$\angle H(j\omega) = -\tan^{-1}(\omega R_oC_o) = -\tan^{-1}(\omega/\omega_p)$$

$$\left|\frac{G_m R_o}{\sqrt{1+\omega^2 R_o^2 C_o^2}}\right| = 1$$

$$\frac{G_m R_o}{\omega_u R_o C_o} \approx 1 \Rightarrow \omega_u = \frac{G_m}{C_o}$$ (unity gain frequency)

$$\frac{A_o}{\omega_u/\omega_p} \approx 1 \Rightarrow \omega_u = A_o\omega_p =$$ DC gain × -3dB bandwidth




$$\left(V_i - \frac{V_o}{k}\right) A(s) = V_o$$

$$\frac{V_o}{V_i} = \frac{k}{1 + \frac{k}{A(s)}} \qquad A(s) = \frac{A_0}{1+s/\omega_p}$$

$$= \frac{k}{1 + \frac{k(1+s/\omega_p)}{A_0}}$$

$$= \frac{k}{1 + \frac{k}{A_0} + \frac{k}{A_0}\frac{s}{\omega_p}}$$

$$= \frac{k}{1+k/A_0} \cdot \frac{1}{1 + \left(\frac{k/A_0}{1+k/A_0}\right)\frac{s}{\omega_p}}$$

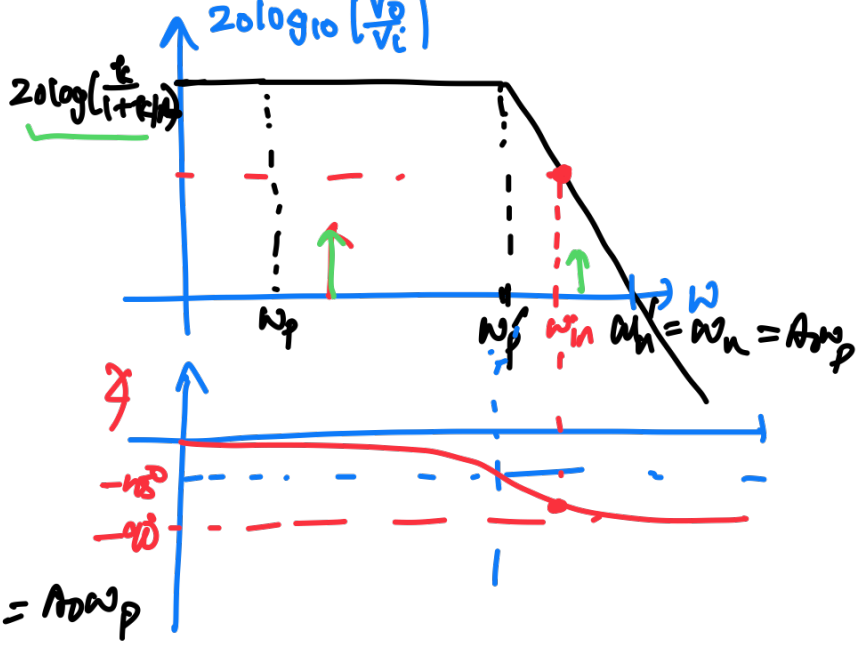
$$\frac{V_o}{V_i} = \frac{k}{1+k/A_0} \cdot \frac{1}{1 + \frac{s}{\omega_p(1+A_0/k)}}$$

DC gain $= \frac{k}{1+k/A_0}$

$$s_1 = -\omega_p(1+A_0/k)$$

$$\omega_p' = \omega_p(1+A_0/k)$$

$$\approx \frac{A_0\omega_p}{k} = \frac{\omega_n}{k}$$

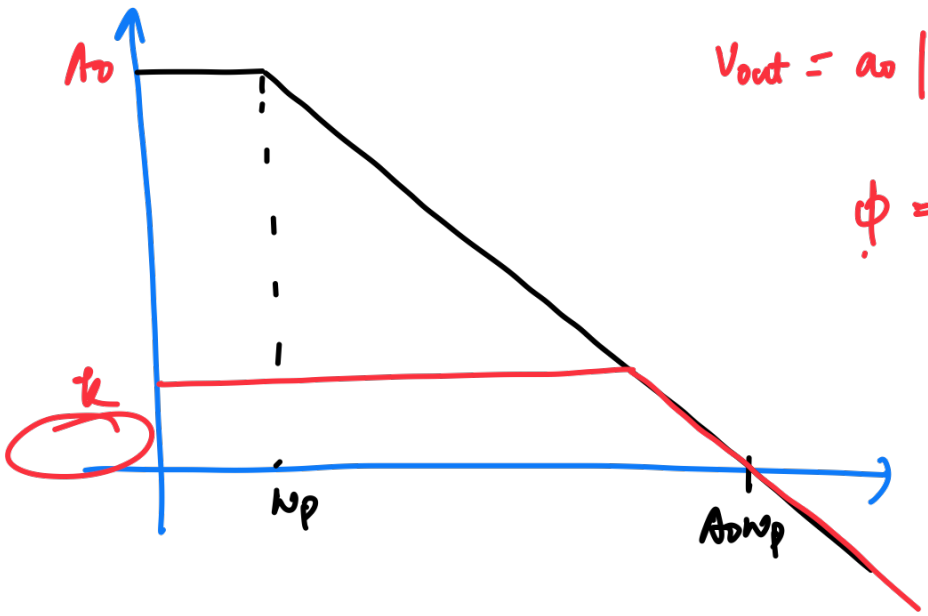


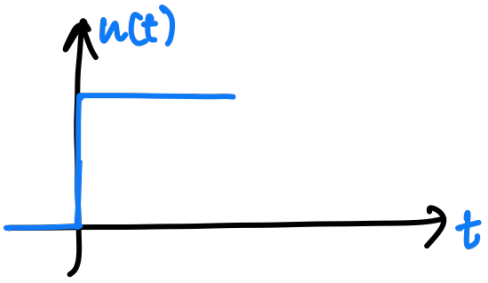

$$\omega_n' = \frac{k}{1+k/A_0} \cdot \omega_p(1+A_0/k) = A_0\omega_p$$

$V_i = a_0 \sin(\omega_{in} t)$

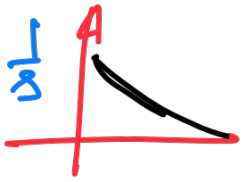
$V_{out} = a_0 \left|\frac{V_o}{V_i}\right|_{\omega=\omega_{in}} \times \sin(\omega_{in} t \pm \phi)$

$\phi = \angle \frac{V_o}{V_i}\Big|_{\omega=\omega_{in}}$




$V_i(t) = u(t) \rightarrow V_i(s) = \frac{1}{s}$



$V_o(t) =$

$V_o(s) = \left(\frac{V_o}{V_i}(s)\right) \times V_i(s)$

closed-loop transfer fn.

$$V_o(s) = \frac{k/(1+k/A_0)}{1+\frac{s}{\omega_p'}\cdots} \times V_i(s)$$

$$= \frac{k}{1+k/A_0}\left[\frac{1}{1+s/\omega_p'} \times \frac{1}{s}\right]$$

$$= \frac{k\omega_p'}{1+k/A_0}\left[\frac{P}{s} + \frac{Q}{s+\omega_p'}\right]$$

with partial fr.

To find P, $s=0$ in $\frac{s}{s(s+\omega_p')} \Rightarrow \frac{1}{\omega_p'} = P$

Q, $s = -\omega_p'$ $\qquad Q = -1/\omega_p'$

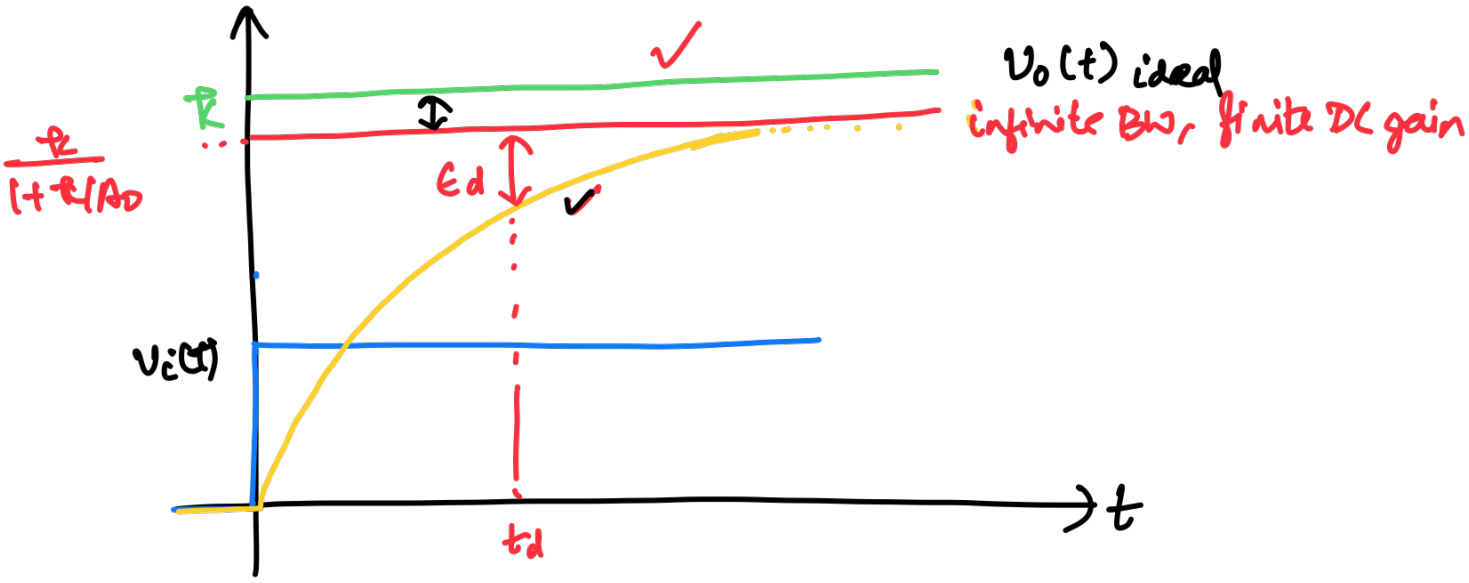
$$V_o(s) = \frac{k}{1 + k/A_0}\left[\frac{1}{s} - \frac{1}{s + \omega_p'}\right]$$

$$v_o(t) = \mathcal{L}^{-1}\{V_o(s)\}$$

$$= \frac{k}{1 + k/A_0}\left[u(t) - e^{-\omega_p' t}u(t)\right]$$

$$= \frac{k}{1 + k/A_0}\left[1 - e^{-\omega_p' t}\right]u(t)$$

$e^{-\omega_p' t} \to 0$ ✓




Steady state error, $\epsilon_s = \lim_{t\to\infty} \dfrac{v_o(t)|_{\text{ideal DC gain}} - v_o(t)|_{\text{real dc gain}}}{\lim_{t\to\infty} v_o(t)|_{\text{ideal dc gain}}}$

$$= \frac{k - \frac{k}{1+k/A_0}}{k} = \frac{k}{A_0}\Bigg|_{A_0\to\infty} = 0$$

$k = 10$, $\epsilon_s < 5\%$

$$\epsilon_d = \frac{\lim_{t=t_d} v_{out}(t)\Big|_{w_p \to \infty, A_0} - v_{out}(t)\Big|_{actual}}{\lim_{t=t_d} v_{out}\Big|_{w_p \to \infty, A_0}}$$

$$= \frac{\frac{1}{1+k}A_0 - \frac{1}{1+k}A_0\left(1 - e^{-w_p' t_d}\right)}{\frac{1}{1+k}A_0}$$

$$= e^{-w_p' t_d}$$

$\epsilon_d < 0.05$

$$e^{-w_p' t_d} < 0.05$$

$$\ln\left(e^{-w_p' t_d}\right) < \ln(0.05)$$

$$-w_p' t_d < \ln(0.05)$$

$$w_p' > \frac{-\ln(0.05)}{t_d} = \frac{\ln(20)}{t_d}$$